The collection of homomorphisms satisfying the conditions of (i) and (ii) is called a **morphism of resolutions**, $f : P_\bullet \to Q_\bullet$.

(iii) Clearly, such a morphism f defines a linear map $\psi_i(P, Q, f) : \mathrm{Tor}_i^P(M, N) \to \mathrm{Tor}_i^Q(M, N)$, where the superscripts P and Q mean that the Tor groups are defined using the resolutions $P_\bullet$ and $Q_\bullet$. Show that the maps $\psi_i(P, Q, f)$ don't really depend on f (so they can be denoted by $\psi_i(P, Q)$).

(iv) Deduce that $\psi_i(P, Q)$ are isomorphisms (use that $\psi_i(P, P) = \mathrm{id}$ and $\psi_i(Q, R) \circ \psi_i(P, Q) = \psi_i(P, R)$).

(v) Define similar maps $\xi_i(Q, P, f) : \mathrm{Ext}_P^i(M, N) \to \mathrm{Ext}_Q^i(M, N)$ and show that they are independent on f and are isomorphisms.

Problem 8.2.6. (i) Show that $\mathrm{Tor}_0(M, N) = M \otimes_A N$ and that $\mathrm{Ext}^0(M, N) = \mathrm{Hom}_A(M, N)$.

(ii) Show that the group $\mathrm{Ext}^1(M, N)$ as defined here is canonically isomorphic to the one defined in Problem 3.9.1.

(iii) Let

$$0 \to N_1 \to N_2 \to N_3 \to 0$$

be a short exact sequence of left A-modules.

Show that there are long exact sequences

$$0 \to \mathrm{Hom}_A(M, N_1) \to \mathrm{Hom}_A(M, N_2) \to \mathrm{Hom}_A(M, N_3) \to$$

$$\mathrm{Ext}^1(M, N_1) \to \mathrm{Ext}^1(M, N_2) \to \mathrm{Ext}^1(M, N_3) \to \mathrm{Ext}^2(M, N_1) \to \dots$$

and

$$\dots \to \mathrm{Tor}_2(M, N_3) \to \mathrm{Tor}_1(M, N_1) \to \mathrm{Tor}_1(M, N_2) \to \mathrm{Tor}_1(M, N_3)$$
$$\to M \otimes_A N_1 \to M \otimes_A N_2 \to M \otimes_A N_3 \to 0.$$

This shows that, as we mentioned above, the Tor and Ext groups "quantify" the extent to which the functors $M \otimes_A$ and $\mathrm{Hom}_A(M, ?)$ are not exact.

Hint: Use Problem 7.8.5.

(iv) Show that $\mathrm{Tor}_i^A(M, N)$ can be computed by taking a projective resolution of N, tensoring it with M, and computing cohomology.